SAT MATH

Algebra • Skill 1.1 • Guided notes

Skill 1.1 • Linear Equations in One Variable

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Solve linear equations, including equations with parentheses, fractions and decimals.
- Decide whether a linear equation has one solution, no solution, or infinitely many solutions.
- Find the value of an expression without solving for x first, and write equations from word problems.

Vocabulary. linear equation, solution, distributive property, least common denominator, no solution, infinitely many solutions.

1. The strategy

Key idea

To solve a linear equation: (1) _____ to remove parentheses; (2) multiply by the _____ to clear fractions; (3) collect the x-terms on one side and the _____ on the other; (4) _____ by the coefficient of x. Check by substituting.

Example 1 • Parentheses

Solve $4(x-3)+5=2x+9$.

Example 2 • Fractions

Solve $\dfrac{x}{3}+\dfrac{x+1}{4}=9$.

2. How many solutions?

Key idea

After simplifying to the form $ax+b=cx+d$:

- if $a \neq c$: exactly _____ solution;
- if $a=c$ and $b \neq d$: _____ solution (the x-terms cancel and a false statement remains, like $5=7$);
- if $a=c$ and $b=d$: _____ solutions (both sides are identical).

Example 3

For which value of k does $3(2x + k) = 6x + 12$ have infinitely many solutions? For which values does it have no solution?

3. Find the expression, not x

The SAT often asks for an expression such as $10x - 4$. Look for a _____ of the given equation.

Example 4

If $5x - 7 = 13$, what is the value of $10x - 4$?

4. Word problems

Example 5

A plumber charges a \$65 call-out fee plus \$40 per hour. A bill was \$245. How many hours did the plumber work?

AP exam tip

[SAT tip] Use Desmos when an equation is messy: graph $y =$ left side and $y =$ right side and read the x-coordinate of the intersection. Parallel lines mean no solution; the same line means infinitely many.

Common mistakes

- Distributing to only the first term: $-2(x - 3) \neq -2x - 3$; it is $-2x + 6$.
- Multiplying only some terms by the common denominator.

Check your understanding

1. $7x - 4 = 3x + 20$ gives $x =$ ____. 2. $2(x + 1) = 2x + 3$ has _____ solution. 3. If $3x + 7 = 22$, then $6x - 1 =$ _____.